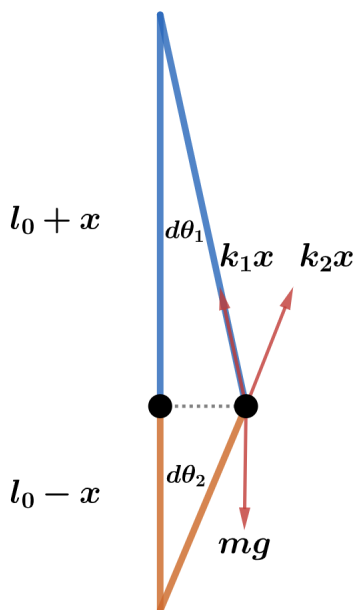


2020A F=ma Exam: Problem 25

Kevin S. Huang



At equilibrium, the mass is at distance x below the middle of the two springs where

$$k_1 x + k_2 x = mg$$

$$x = \frac{mg}{k_1 + k_2}$$

For small horizontal perturbations, we require the force to be restoring for a stable equilibrium,

$$k_1 x \sin(d\theta_1) > k_2 x \sin(d\theta_2)$$

Keeping to leading order,

$$k_1 x \left(\frac{\delta x}{l_0 + x} \right) > k_2 x \left(\frac{\delta x}{l_0 - x} \right)$$

$$k_1(l_0 - x) > k_2(l_0 + x)$$

$$l_0(k_1 - k_2) > x(k_1 + k_2)$$

Thus,

$$k_1 - k_2 > \frac{mg}{l_0}$$

so the answer is C.